

WHAT TO LOOK FOR IN AIR PURIFIERS

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There is a general perception that a house provides a protective shield from outdoor toxic air pollution. But very few people know that indoor air is as bad as outdoor air. Indoor air is, in fact, much more polluted. Studies indicate that carbon-dioxide and various gases in the form of volatile organic compounds are present indoors as the main pollutants, which are hazardous. These pollutants get much more concentrated inside the home.

On an average, we spend approximately eighty per cent of our time indoors. And what is worse is that human eyes can only see particles larger than ten microns. You can imagine the lethal effects of the invincible air-borne particles present in the air. Therefore indoor air pollution is a major cause for concern as it breeds a host of problems including such diseases as lung and heart, mental health problems and so on. Considering the adverse impact of pollution on health and environment, an air purifier is no more a luxurious indulgence but a requisite to lead a healthy life.

Types of air purifiers

The core purpose of air purifiers is to cleanse the air. These are available in various shapes, sizes and colours; however, not all work the same way. Air purifiers are equipped with different technologies and features and, hence, effectiveness varies from one product to another.

HEPA technology. High efficiency particulate air (HEPA)-based air filters are the most efficient for air purifiers.

Usually made of fibreglass, HEPA filters have a standard trapping of at least 99.7 per cent, which effectively removes particles as small as 0.3 microns.

These filters typically last two to four years. This technology has an extra edge in trapping harmful particles in comparison with other air purifiers. These are effective in removing dust, pollen, dust mites, pet dander, mould and more. With HEPA-based air purifiers, these unhealthy particles hardly get released back into the air and also do not generate ozone gas.

The only flip side is that these cannot remove chemicals, odours, gases, germs and viruses (0.01 microns and below) successfully. Nonetheless, these are the most sought-after when it comes to air purifiers.

Activated carbon air filters. These have many small pores between carbon atoms that are highly absorbent with good chemical bonding. These are effective in trapping gases, chemical emissions, odours and smoke. The best part about activated carbon air filters is that these can curb pollutants from contaminating the atmosphere by not releasing them back into the air.

Further, these do not generate any hazardous byproducts or ozone gas. However, these filters get affected with time as contaminants get accumulated or trapped on the



Air purifier AC5659 by Philips